

USER GUIDE

From one drawing to a moving laser

DWG to ESP300 turns a drawing into laser stage motion, focuses the beam automatically with a camera, and lets you watch the real machine as it runs — all from one screen. This guide takes you through every step, start to finish.

GETTING STARTED

What does this app do?

It loads a drawing, lets you preview it, and runs the real machine — all in one window.

1 Load a drawing

Start by opening a drawing file (DWG or DXF) you already have.

2 Preview it on screen

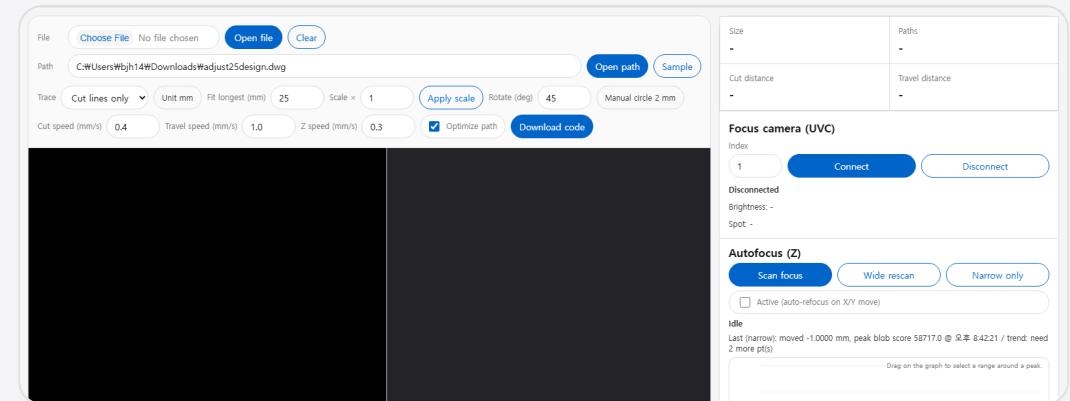
See exactly how it will move, in both 2D and 3D, before anything happens.

3 Let the camera focus it

The connected camera finds the sharpest position on its own.

4 Run it on the real machine

Once you're happy with the preview, run the same plan for real.



GETTING STARTED

Key features

Four things that make this app worth using.



Your drawing becomes motion

No separate conversion step — just open the file you already have and it turns straight into a runnable path.



Focus is handled for you

The camera sweeps through Z on its own and automatically settles on the brightest, sharpest spot.



Everything, one screen

Previewing the drawing, watching the camera, moving the stage, and running the job never leave this one window.



Safe to practice in

A preview-only practice mode is on by default, so you can try things out before anything really moves.

FIRST LAUNCH · STEP 01

Before you start

One icon on the desktop is all it takes to launch the app.

1 Double-click "DWG to ESP300 열기" on the desktop

The app starts, and a browser window opens on its own a moment later.

2 Give it a few seconds

The very first launch can take a little longer than usual.

3 Already running? No need to relaunch

Clicking the icon again just brings you back to the window that's already open.

- **Good to know.** If the computer goes to sleep, the connection to the machine can drop. Turn off sleep mode while you're working.



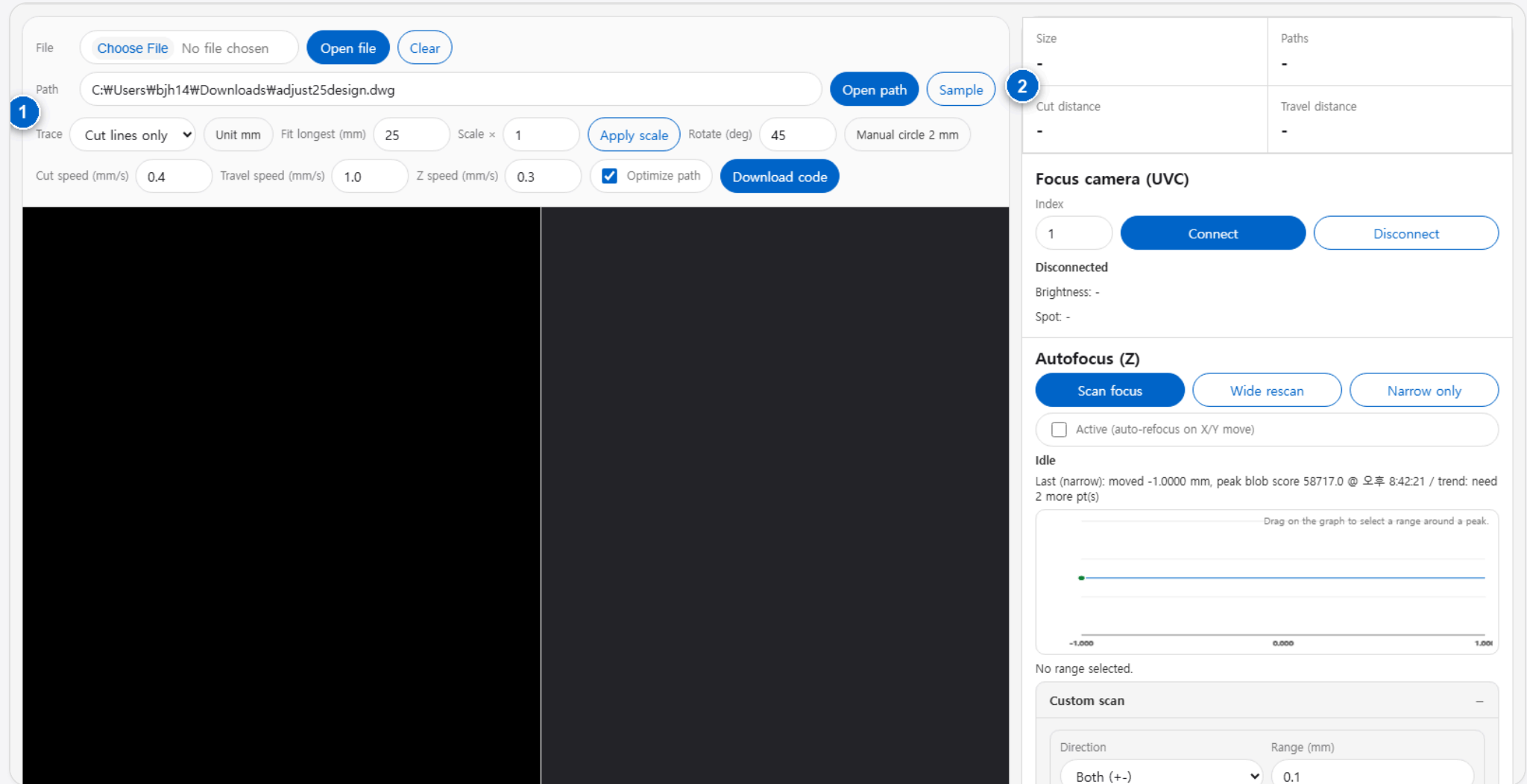
DWG to ESP300 열기

Desktop shortcut

MAIN SCREEN · STEP 02

The main screen at a glance

The screen splits into two halves: your workspace on the left, the tool panel on the right.



1 Workspace

Where you open a drawing, check it, and play it back.

2 Tool panel

The camera view, focus controls, and machine controls all live here.

WORKSPACE · STEP 03

Load a drawing

Open a drawing and set the basics from the toolbar at the top.

The screenshot shows the top toolbar of the DWG to ESP300 application. It includes a 'File' section with 'Choose File', 'No file chosen', 'Open file', and 'Clear' buttons. A 'Path' input field contains the file path 'C:\Users\Wbjh14\Downloads\Wadjust25design.dwg', with 'Open path' and 'Sample' buttons. A 'Trace' section has a dropdown menu set to 'Cut lines only'. Other settings include 'Unit mm', 'Fit longest (mm)' set to 25, 'Scale x' set to 1, 'Apply scale' button, 'Rotate (deg)' set to 45, and 'Manual circle 2 mm'. At the bottom, there are input fields for 'Cut speed (mm/s)' (0.4), 'Travel speed (mm/s)' (1.0), 'Z speed (mm/s)' (0.3), a checked 'Optimize path' checkbox, and a 'Download code' button. Five blue circles with numbers 1 through 5 are overlaid on the interface to indicate the sequence of steps: 1 on 'Open file', 2 on the 'Path' input, 3 on the 'Sample' button, 4 on the 'Cut lines only' dropdown, and 5 on the 'Download code' button.

1 Open a file

Click "Open file" and choose a drawing from your computer.

2 Open by path

You can also type in the file's location directly.

3 Try the sample first

Click "Sample" to practice with a drawing that's already set up.

4 Choose which lines count

"Cut lines only" automatically picks out just the lines meant to be cut.

5 Download the result

"Download code" saves exactly the plan you just previewed as a file.

WORKSPACE · STEP 04

Check it in the preview

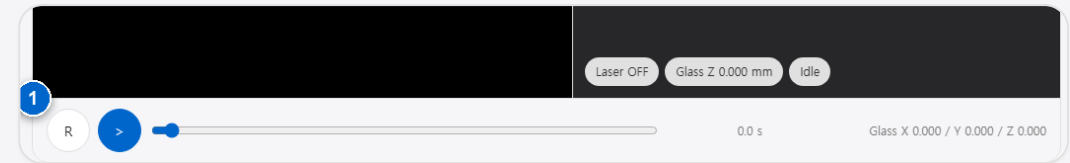
Before anything runs for real, see how it will move right on screen.

1 Play the motion back

Press ► to play the path in real order. Drag the bar underneath to jump straight to any point.

The left side is a top-down 2D view; the right side looks like the real machine in 3D. Drag with your mouse on the 3D view to rotate it to any angle.

- **Good to know.** The bar at the bottom shows the laser on/off state, the current height, and progress, all updating live.



TOOL PANEL · STEP 05

Watch the camera live

See the connected camera's feed right here.

1 Focus camera

This is where the camera feed appears.

2 Index number

Which camera to connect to. If you only have one, leave the default as is.

3 Connect

Click to connect the camera. Once connected, the live feed appears here.

4 The video itself

Click anywhere on the video to set that spot as the target.

Size -	Paths -
Cut distance -	Travel distance -

1

2

3

Focus camera (UVC)

Index
1

Connect **Disconnect**

Disconnected

Brightness: -

Spot: -

Autofocus (Z)

Scan focus **Wide rescan** **Narrow only**

☐ Active (auto-refocus on X/Y move)

Idle

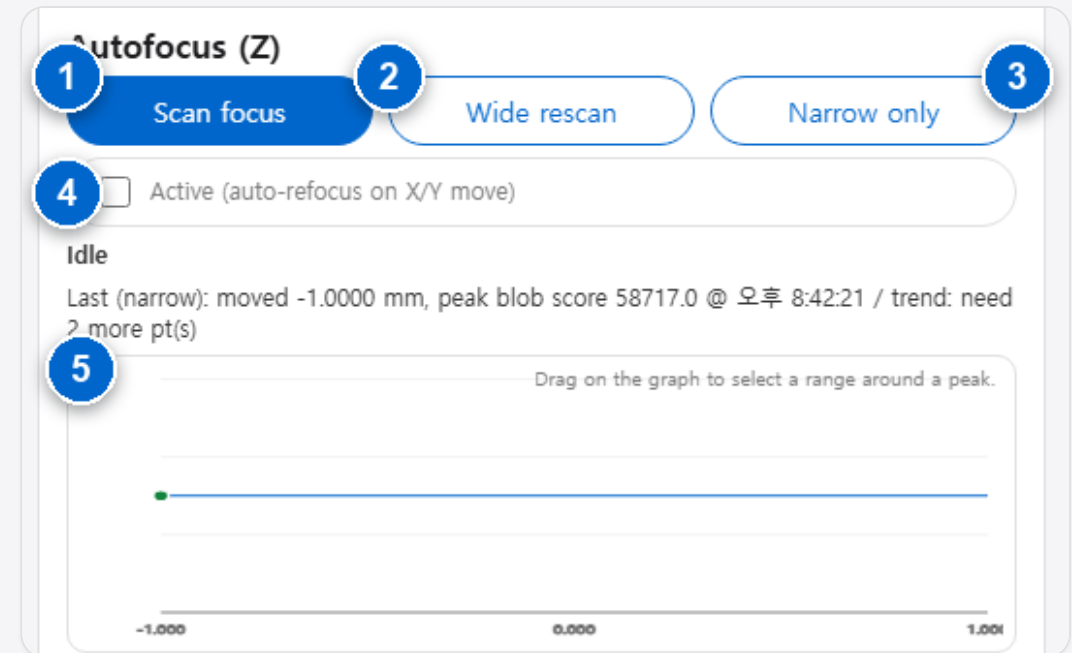
Last (narrow): moved -1.0000 mm, peak blob score 58717.0 @ 오후 8:42:21 / trend: need

TOOL PANEL · STEP 06

Let it focus automatically

One click finds the sharpest height for you.

- 1 Scan focus**
The standard way to focus. Use this right after connecting.
- 2 Wide rescan**
Looks over a wider range. Use this if focus seems way off.
- 3 Narrow only**
A quicker recheck over a small range.
- 4 Active**
Turn this on and focus is automatically re-checked every time you move to a new spot.
- 5 The chart**
Shows which height was brightest. Drag across it to re-check just that range.



TOOL PANEL · STEP 07

Fine-tune the focus further

For when the basic autofocus isn't quite enough. Click to expand or collapse each one.

1 Custom scan

Set your own range and step size to examine one specific window closely.

2 Timing check

A self-check that confirms your current settings are accurate enough.

3 Spiral survey

Visits several points in turn, checking focus at each and building a map.

- **Be careful.** Keep an eye on the screen while it visits multiple points. You can always press `Space` to stop it immediately if anything looks wrong.

The screenshot shows a software interface with three expandable sections, each indicated by a numbered blue circle (1, 2, 3) on the left.

- Section 1: Custom scan**
 - Direction: Both (+-) (dropdown menu)
 - Range (mm): 0.1 (input field)
 - Step (mm): 0.005 (input field)
 - Settle (s): 0.05 (input field)
 - Run custom scan (blue button)
- Section 2: Timing check**
 - Verify timing (slow vs fast) (blue button)
 - No timing check yet. (text)
- Section 3: Spiral survey**
 - ☐ Narrow only (skip wide, even on point 1) (checkbox)
 - Survey spiral path (blue button)
 - Stop survey (blue button)
 - Survey done: 1/5 points (text)
 - X/Y map fills in live as the survey runs. (text)

TOOL PANEL · STEP 08

Connect the machine

Before moving the real machine, you need to connect to it first.

- 1 Jog tab**
Where you drive the machine directly. Open by default.
- 2 Shapes tab**
Draw your own shapes without a drawing file.
- 3 Paths tab**
Manage each loaded path one by one.
- 4 Port**
The communication port the machine is connected through. Leave the default unless you've been told otherwise.
- 5 Travel range**
The farthest the machine is allowed to move. If it ever tries to go past this, it stops automatically for safety.
- 6 Dry run (practice mode)**
While checked, nothing really moves — you only see it on screen. Uncheck it when you're ready for real motion.
- 7 Connect**
Click to connect to the machine.

The screenshot shows a software interface for controlling a machine. At the top, there are three tabs: 'Jog' (highlighted with a blue underline and a blue circle with the number 1), 'Shapes' (with a blue circle with the number 2), and 'Paths' (with a blue circle with the number 3). Below the tabs is a section titled 'Manual jog (ESP300)'. Inside this section, there are several controls: a 'Port' dropdown menu showing 'COM5' (with a blue circle with the number 4), a 'Baud' rate dropdown showing '19200', a 'Y range (mm)' input field showing '75' (with a blue circle with the number 5), a 'Z range (mm)' input field showing '20', a checkbox labeled 'Dry run (no hardware)' which is checked (with a blue circle with the number 6), and two large buttons at the bottom: 'Connect' (highlighted with a blue circle with the number 7) and 'Disconnect'.

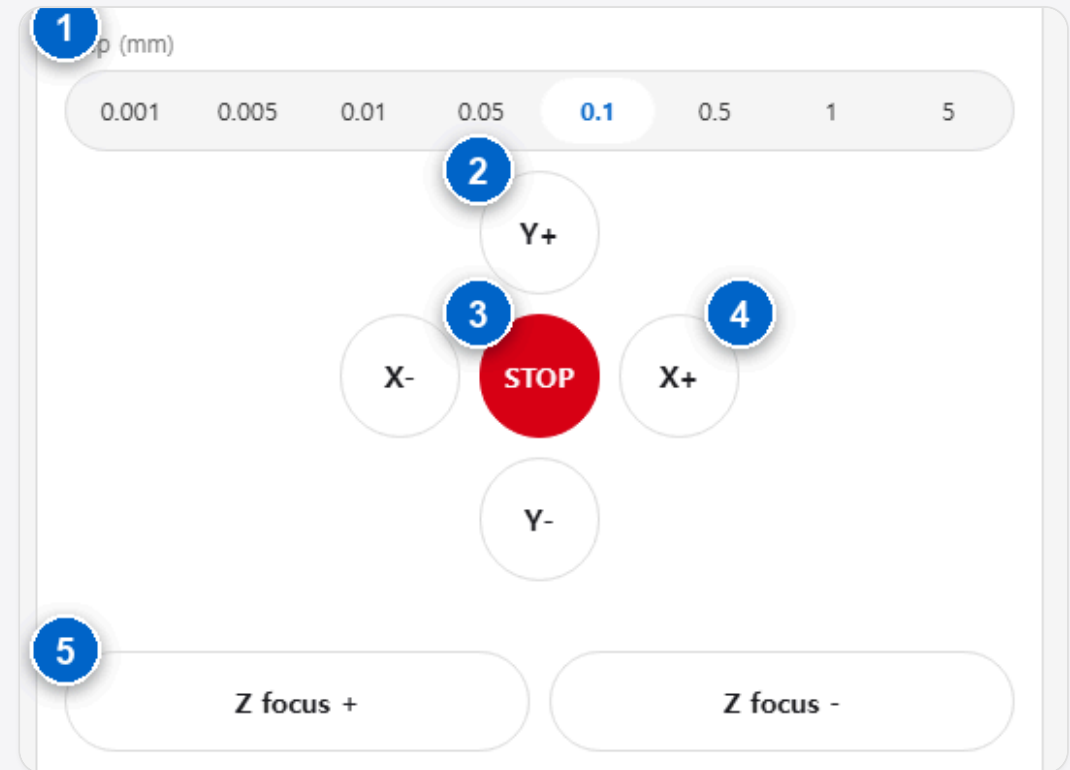
TOOL PANEL · STEP 09

Move it by hand

Once connected, the arrow buttons let you drive the machine yourself.

- 1 Distance per press**
Choose how far one press moves. Smaller numbers mean smaller steps.
- 2 Direction buttons**
Hold one down and it keeps moving that way. The arrow keys on your keyboard do the same thing.
- 3 Emergency stop**
The red STOP button in the middle. It stops instantly, and you'll need to connect again before moving further. The `Space` key does the same thing.
- 4 Focus height**
The Z focus buttons nudge the height (focus) up or down a little at a time.

- **Be careful.** It's always safe to hit the emergency stop. It's built to stop instantly, no confirmation needed.



TOOL PANEL · STEP 10

Set a home position, then run

Save the spots you use often, and run for real once you're ready.

- 1 Home**
Moves to the machine's original reference point (0,0,0). This can be a large move, so it asks you to confirm first.
- 2 Set local home**
Saves the current spot as your own reference point. The real machine doesn't move.
- 3 Go to local home**
Returns to the reference point you saved.
- 4 Go to coordinates**
Type in the position you want (X/Y/Z) and press "Go".
- 5 Run on machine**
Runs the plan you've been previewing on the real machine. Asks you to confirm before it starts.
- 6 Stop run**
Stops a run that's already in progress.

The screenshot shows a control panel with the following elements:

- Step 1:** A button labeled "Home (0, 0, 0)".
- Step 2:** A button labeled "Set local home (here = 0)".
- Step 3:** A button labeled "Go to local home".
- Step 4:** A section titled "to (mm)" containing three input fields with values "0.996", "11.000", and "21.214", followed by a blue "Go" button.
- Coordinate Display:** Below the inputs, it shows "X 0.996 / Y 11.000 / Z 21.214 (focus trim -1.000)" and "74.00mm to XY limit / 20.50mm of Z headroom".
- Instructions:** "Hold a button to jog continuously. Arrow keys move X/Y, Page Up/Down move Z, Space is E-STOP."
- Step 5:** A section titled "Run planned path on machine" containing a large blue "Run on machine" button.
- Step 6:** A button labeled "Stop run".
- Status:** At the bottom left, it says "Idle".

TOOL PANEL · STEP 11

Draw shapes without a drawing

No drawing file? You can build your own shape directly.

- 1 Pick a shape**
Choose from a line, rectangle, circle, or spiral.
- 2 Enter size and position**
Type in numbers for whichever shape you picked.
- 3 Blank**
Clears every shape you've drawn so far and starts fresh.
- 4 Add shape**
Adds one shape using the values you just entered.
- 5 Delete selected**
Removes whichever shapes are checked in the Paths tab.

The screenshot shows the 'Manual shapes' tool panel. It features a tabbed interface with 'Line', 'Rect', 'Circle', and 'Spiral'. The 'Line' tab is active. Below the tabs are four input fields: the first is labeled '0', the second 'Y1' with '0', the third 'X2' with '10', and the fourth 'Y2' with '0'. At the bottom are three buttons: 'Blank', 'Add shape', and 'Delete selected'. Numbered callouts are placed as follows: 1 on the 'Line' tab, 2 on the first input field, 3 on the 'Blank' button, 4 on the 'Add shape' button, and 5 on the 'Delete selected' button.

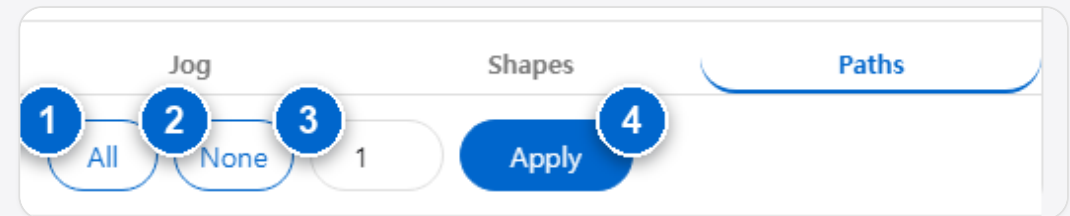
TOOL PANEL · STEP 12

Choose and manage paths

When there's more than one path loaded, pick just the ones you want.

- 1 All**
Selects every path at once.
- 2 None**
Clears the selection.
- 3 Pass count**
How many times to go over the same spot.
- 4 Apply**
Applies that pass count to every path you've selected, all at once.

Below this, each path is listed one at a time — you can check individual ones or change their pass count separately, too.



SETTINGS · STEP 13

A look at the settings

There's no separate "Settings" screen — you type numbers in right next to each feature. Here's a quick reference for the ones you'll see most.

NAME	WHERE TO FIND IT	WHAT IT CONTROLS	DEFAULT
Cut speed	Load-a-drawing screen	How fast it moves while actually cutting	0.4mm/s
Travel speed	Load-a-drawing screen	How fast it moves when not cutting	1.0mm/s
Z speed	Load-a-drawing screen	How fast the focus height changes	0.3mm/s
Travel range (X-Y)	Connect-the-machine screen	The farthest it can move side to side / front to back	75mm
Travel range (Z)	Connect-the-machine screen	The farthest it can move up and down	20mm
Distance per press	Move-it-by-hand screen	How far one button press moves it	0.1mm

● **Good to know.** The travel range values are there for safety. Unless you have a specific reason to change them, it's best to leave the defaults as they are.

THE ONE FLOW YOU'LL USE MOST

The whole thing in five steps

Remember these five and you're set.

01

Load

Open a drawing from the toolbar.

>

02

Preview

Press play to check the motion.

>

03

Connect

Connect the camera and the machine.

>

04

Focus

Let Scan focus handle it automatically.

>

05

Run

Run it for real with Run on machine.

GOOD TO KNOW

Tips for working faster

1 Leave Active turned on

It re-focuses automatically every time you move, so you don't have to click it yourself.

2 Start in practice mode

Get comfortable with Dry run turned on first, then turn it off once you're ready to move for real.

3 The keyboard is often faster

Arrow keys move it, Page Up/Down changes height, and Space stops everything instantly.

4 Practice with the sample first

No drawing ready yet? Use the sample drawing to get the whole flow down first.

5 Don't let the computer sleep mid-job

Sleep mode can drop the connection.

6 When in doubt, stop first

If you're ever unsure, hit STOP (emergency stop) first and figure out what's going on. There's nothing you can't undo by stopping.

TROUBLESHOOTING

If this happens, try this

Q. Nothing shows up in the camera view

The camera probably isn't connected yet. Click Connect in the Focus camera panel first.

Q. The machine isn't moving

If Dry run (practice mode) is checked, nothing actually moves. Uncheck it and connect again.

Q. I see a message about the travel range being exceeded

It means the move would have gone past the safe travel range that's set. If you really do need to reach that position, increase the travel range and connect again.

Q. I hit the emergency stop — what now?

Pressing STOP disconnects the machine for safety. Once you've checked everything is okay, click Connect again to keep going.

Q. Focus is taking a really long time

There may be a problem with the camera connection. Watch the screen, and if something looks off, press STOP and reconnect the camera.

Q. The run stopped suddenly

Check the connection to the machine. Inspect the cable or USB connection, then reconnect and run again.

CLOSING

You're ready to go

From loading a drawing, to focusing with the camera, to running the real machine — just follow the order laid out in this guide. If you ever get stuck, flip back to the matching page any time.